LESSON 8 • TERM 1 • LEVEL 1

# Balance and Swap

Addition and subtraction: the equals sign

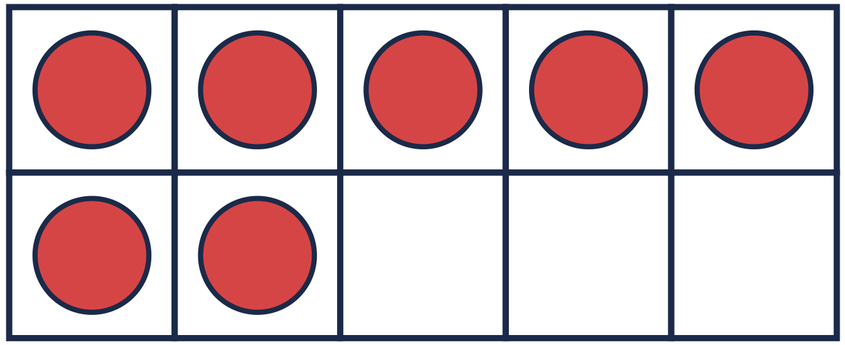
**TODAY WE WILL**

Balance two sides, and find missing numbers by giving and taking.



# Warm-up: fingers to ten

2–8 MIN



I show 7. How many more fingers make 10?

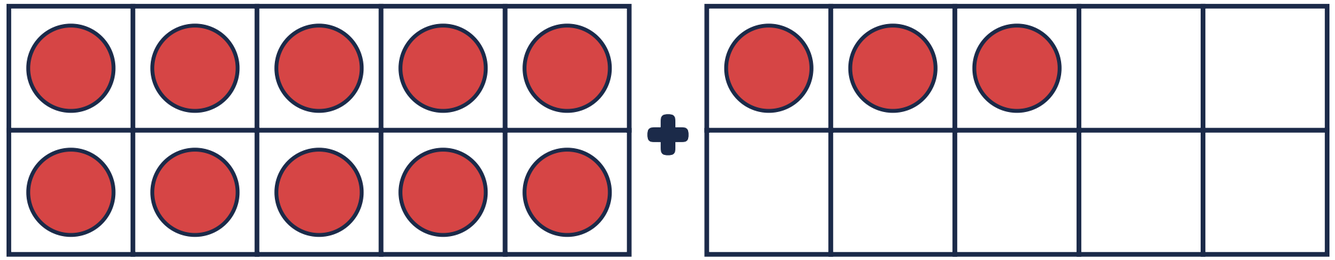
**7** 4 **9** 6 **2**

Show me your fingers!



# Ten and more: how many to twenty?

2–8 MIN



Start at 13. How many more make 20?

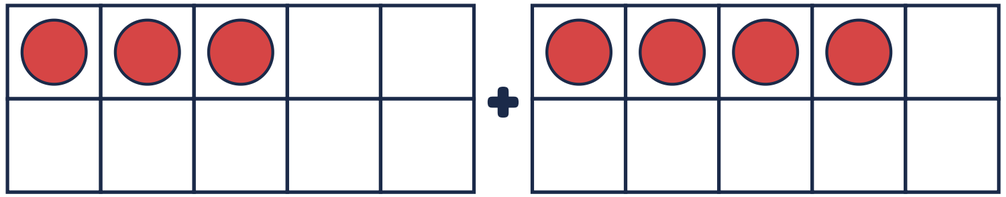
**13** **16** **18** **15**

# The equals sign

2–8 MIN

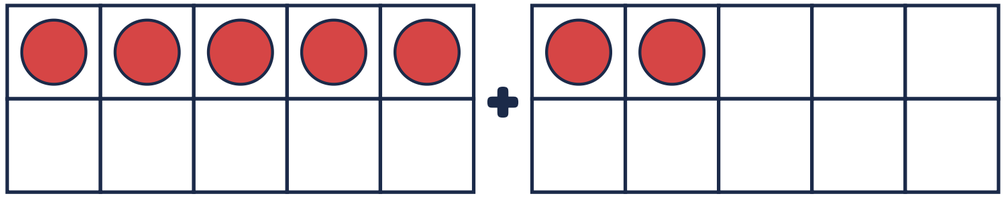
**LEFT SIDE**

3 + 4



**RIGHT SIDE**

5 + 2



**= means the same amount as**

**Is 3 + 4 the same amount as 6 + 2?**

# Sneak-peek puzzles

8–12 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

## PUZZLE 1: anyone can try

Ravi has 6 red beads and 4 blue beads.

Siti has 5 red beads and 5 blue beads.

Do they have the same number of beads? Draw dots to help, then circle Yes or No.

## PUZZLE 2: think

8 + 5 = 9 + ___

What number goes in the blank? Write how you know.

# Explore: two plates

12–28 MIN

1. Put your two plates side by side: LEFT and RIGHT.
2. Left plate: a pile of 6 and a pile of 4.
3. Right plate: a pile of 5, and a second pile.
4. Make the plates match. How many in the second pile?

**WHAT IF?**

What if the left plate has 7 and 3? What if it has 9 and 2?

12–28 MIN

# Give and take

Left pile 7, right pile 5. Count all. Move 1 across. Count all again.

**7 + 5** | 6 + 6 | **5 + 7** | 4 + 8

Next, match 8 + 4 with 9 + a second pile.

# What did we notice?

12–28 MIN

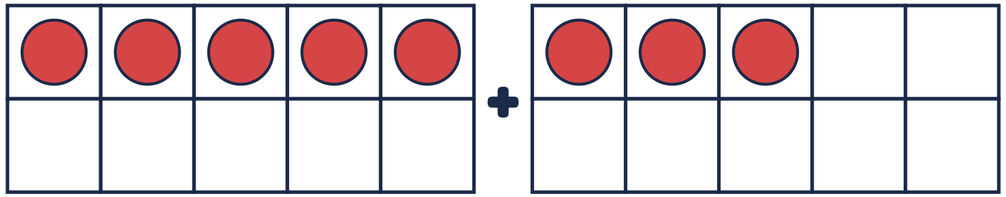
- When one number went up, what happened to the other?
- Did the total change when we gave one across?
- How could you find the second pile without counting everything?

# Equals means balance

28–38 MIN

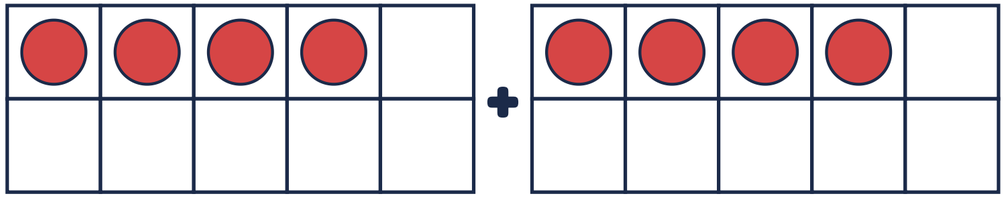
**LEFT SIDE**

5 + 3



**RIGHT SIDE**

4 + 4

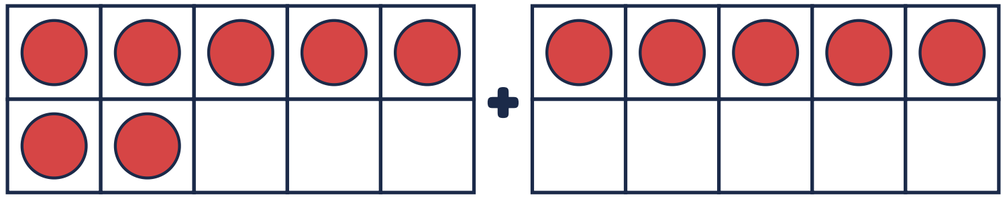


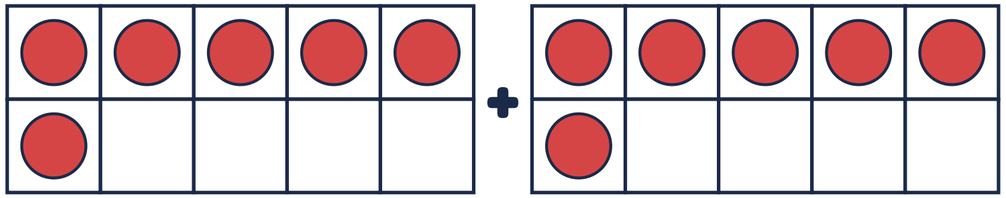
**5 + 3 = 4 + 4. Both sides make 8.**

# Give and take

28–38 MIN

| What we do | What we write |
| --- | --- |
| **Start** | 7 + 5 = 12 |
| **Take 1 from the 7, give it to the 5** | 6 + 6 = 12 |
| **Again** | 5 + 7 = 12 |





**The total stays the same.**

28–38 MIN

# Your turn: the missing number

Write on your paper. About 2 minutes

8 + 6 = 7 + ____

9 + 4 = 10 + ____

6 + 5 = 4 + ____

28–38 MIN

# Check: the missing number

**8 + 6 = 7 + 7**

**9 + 4 = 10 + 3**

**6 + 5 = 4 + 7**

# Bigger numbers, same idea

28–38 MIN

25 + 18 = 27 + ___

- 25 went up to 27: 2 more.
- So the blank is 2 less than 18.
- The blank is 16.

**Check: 25 + 18 = 43 and 27 + 16 = 43.**

# Same total? Do not add yet

28–38 MIN

**SAME TOTAL**

47 + 18 and 45 + 20

47 goes down 2. 18 goes up 2.

**NOT THE SAME**

29 + 14 and 30 + 15

29 goes up 1. 14 goes up 1.

**One number goes down and one goes up by the same amount. The total stays the same.**

28–38 MIN

# Quick-fire: is it balanced?

**6 + 5 = 7 + 4**

**8 + 3 = 9 + 3**

**20 + 14 = 22 + 12**

Thumbs up: balanced. Thumbs down: not balanced.

38–42 MIN

# See-saw stand-up

Arms out wide like a see-saw. Left hand and right hand show the fingers.

6 and 4 | 5 and 5 | 7 and 3

Give one across. Still ten fingers! Sip some water.

# Workbook page 1: Start here

42–70 MIN

Questions 1 to 3. About 8 minutes

**S1**

**Fill in the missing numbers.**

(a) 5 + 3 = 4 + ☐

(b) 6 + 4 = 5 + ☐

(c) 7 + 3 = ☐ + 7

(d) 2 + 8 = 8 + ☐

**S2**

**Mei puts 6 buttons and 2 buttons on the left plate. On the right plate she puts 7 buttons and some more. Draw the right plate in the box so both plates are the same.**

She adds ☐ more buttons

**S3**

**Siti writes 6 + 3 = 5 + 9.**

(a) Is it balanced? Circle Yes or No.

(b) Fill in the missing number: 6 + 3 = 5 + ☐

# Check page 1

42–70 MIN

| | |
|---|---|
| S1 | 4, 5, 3, 2 |
| S2 | 1 more button (6 + 2 = 8 and 7 + 1 = 8) |
| S3 | No. The blank should be 4 (6 + 3 = 9 and 5 + 4 = 9). Siti wrote the total of the left side, as if = meant the answer comes next. |

# Workbook page 2: Start here

42–70 MIN

Questions 4 to 7. About 9 minutes

**S4**

**7 + 5 = 12. Fill in the missing numbers.**

(a) 6 + ☐ = 12

(b) 5 + ☐ = 12

(c) 4 + ☐ = 12

**S5**

**Fill in the missing numbers.**

(a) 20 + 30 = 40 + ☐

(b) 50 + 20 = 60 + ☐

(c) 40 + 20 = 30 + ☐

(d) 30 + 30 = 20 + ☐

**S6**

**Mei has 9 red pens and 6 blue pens. Ben has 10 red pens and some blue pens. They have the same number of pens. How many blue pens does Ben have?**

Ben has ☐ blue pens.

**S7**

**Fill in the missing number.**

35 + 12 = ☐ + 2

# Check page 2

42–70 MIN

| | |
|---|---|
| S4 | 6, 7, 8 |
| S5 | 10, 10, 30, 40 |
| S6 | 5 blue pens (9 + 6 = 10 + 5) |
| S7 | 45 (35 + 12 = 47 and 45 + 2 = 47) |

42–70 MIN

# Choose your next page

| Page | What it is |
| --- | --- |
| **Think harder (page 3)** | Same total without adding |
| **Go further (page 4)** | Two blanks and chains |
| **Investigation (page 5)** | Keep it the same |
| **Check** | Finish and check your own work |

**Everyone may choose. You can swap at any time.**

# Talk and explain

70–76 MIN

A CHILD SAYS

6 + 4 = 10. I add 1 to the 6 and 1 to the 4.

7 + 5 is still 10.

Do you agree? Use your plates to explain.

I think … because …

# Back to the sneak-peek

70–76 MIN

8 + 5 = 9 + ___

**The blank is 4. 9 is 1 more than 8, so the blank is 1 less than 5.**

# Exit ticket

76–82 MIN

On your own. No looking back. About 3 minutes

**E1** **Fill in the missing number.**

9 + 5 = 10 + ▢

**E2** **Is 28 + 6 = 30 + 4 balanced?**

Circle Yes or No.

**E3** **Fill in the missing number.**

30 + 12 = 40 + ▢

# Check the exit ticket

76–82 MIN

| | |
|---|---|
| E1 | 4 |
| E2 | Yes |
| E3 | 2. The 30 went up by 10 to 40, so the 12 goes down by 10 to 2. |

# Look-back

82–85 MIN

**Three minutes. There are no right answers. You can pass. Nobody reads it out.**

| | |
|---|---|
| **Something I did well** | write or draw one thing |
| **The hardest part was …** | ☐ knowing what = means ☐ finding the blank ☐ how the numbers moved<br>☐ the big numbers (Think harder) ☐ something else |
| **What helped me** | ☐ two plates and small things ☐ drawing it ☐ trying again<br>☐ I asked for help ☐ I was not stuck |
| **Next time I will try** | ☐ drawing it ☐ using objects ☐ checking my answer<br>☐ asking a question |
| | ☐ I am still stuck (I will ask in the class chat or show my tutor on camera)<br>☐ I would like to pass |

**Take your time. You can pass.**



85–90 MIN

# Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

Draw and Measure. Bring a cm ruler, 10 paper clips, a pencil and paper. No ruler? Tell your tutor.



OPTIONAL

# Go further: what do you notice?

20 = 11 + ____

20 = 12 + ____

20 = 13 + ____

Keep going. What do you notice about the blanks?

OPTIONAL

# If we have time: pick one

| Activity | What we do |
| --- | --- |
| **Is it always true?** | Swap the two numbers in a sum. Does the total stay the same? |
| **Make a puzzle** | Write a missing-number puzzle for your grown-up |
| **Teach it back** | Show how to find the blank in 25 + 18 = 27 + ___ |

**Optional. Everyone may join.**